MATH 220/199 Week 9 Day 2

Reminders

1. Office hours are held in Davenport 330 at 3pm on Thursdays.
2. Do the Merit Quiz on the Merit Canvas page.

Functions that are Fun to Sketch

As accurately as you can make a sketch of the following three functions:

1. $f(x) = \sin(x) + x$ (HINT: What transformation can be applied to $\sin(x)$ to obtain $f(x)$?)

2. $g(x) = (x^2 + 1)^{-2/3} + (x^2 + 1)^{2/3}$

3. Use your Algebra 2 skills to graph $h(x) = x^4 - 8x^2 + 16$, then use Calculus to make sure everything makes sense.

Function Sketching, but Backwards

For each of the following functions, sketch a graph which has that function as it's derivative. (This will be cool to know in Week 12-ish).

1. $f(x) = |x - 2|$

2. $f(x) = |x|^{1/3}$

3. $f(x) = \begin{cases} \vdots & \vdots \\ -1 & -1 \leq x < 0 \\ 0 & 0 \leq x < 1 \\ 1 & 1 \leq x < 2 \\ \vdots & \vdots \end{cases}$